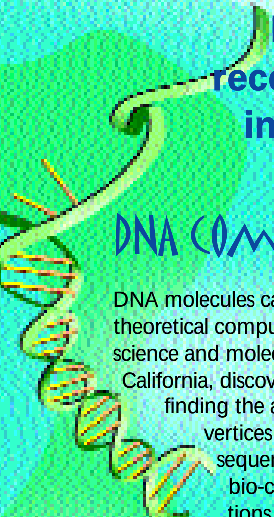
Experience is the best teacher, but not in technology, until recently. Find out how a 250-year-old theorem is being employed in today's systems to make use of the knowledge of the past



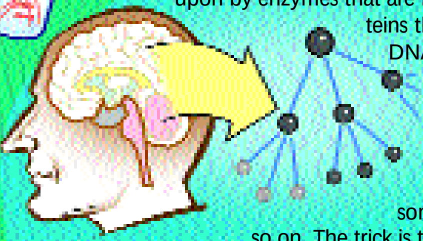
DNA COMPUTING

DNA molecules can compute, as Dr Leonard Adleman, theoretical computer scientist and professor of computer science and molecular biology at University of Southern California, discovered in 1994. His experiment involved finding the answer to a graph theory problem: The vertices of the graph were encoded as DNA sequences in a test tube, and massively parallel bio-chemical reactions resulted in combinations of these sequences. These combinations translated to the huge number of paths through the graph. The problem was to find a particular, 'correct' path.

Adleman was actually able to find it, by decoding the resulting sequences according to the laws of biochemistry. The DNA strands had behaved like a parallel-processing computer.

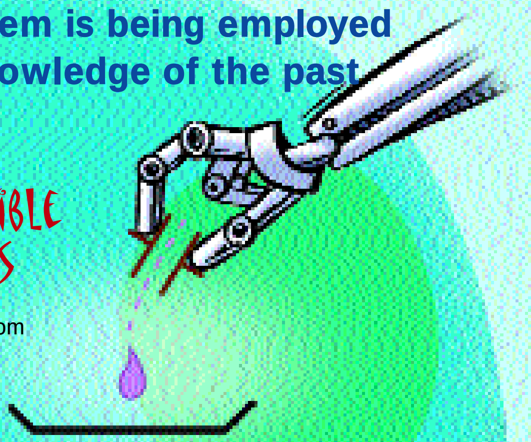
Research expanded the scope of DNA computers to a broad class of problems. The strands in a DNA molecule are much more densely packed than the processing units in a regular computer. In fact, one millilitre of DNA can theoretically perform 10^{22} operations per second. And that's where the potential of DNA computing lies—massive parallelism.

Here's how it works: In a living cell, DNA is operated upon by enzymes that are made up of proteins that 'understand' DNA. These proteins can operate on DNA—cut a strand away, make a copy, paste a strand somewhere else, and so on. The trick is to manipulate these operations. So, just as regular computers or calculators add, subtract and so on, DNA computers have their own set of operations—cutting, copying, etc. Thousands of these operations occur at the same time in a test tube, and so the computation is massively parallel.

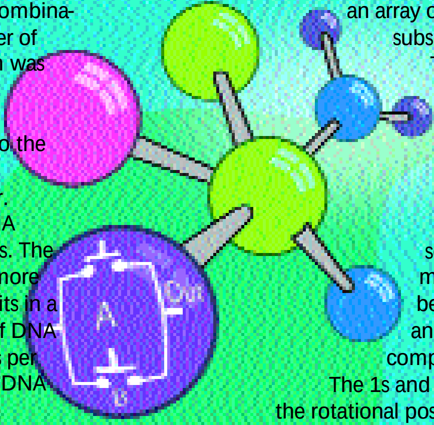


Here's Artificial Intelligence from natural resources: Dr David Stenger of the US Naval Research Laboratory, and Dr James Hickman of Science Applications International Corporation, are linking actual, living neurons into a bio-electronic computer. Drs Stenger and Hickman believe that this bio-electronic mutant might solve pattern-recognition problems. Humans are good at pattern recognition, and it's their neurons that do the job—so why not get them to do something on their own?

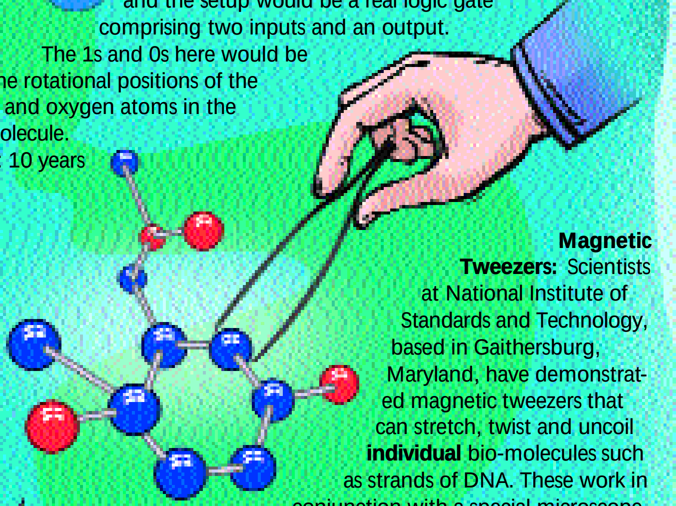
RECENT, INCREDIBLE DEVELOPMENTS



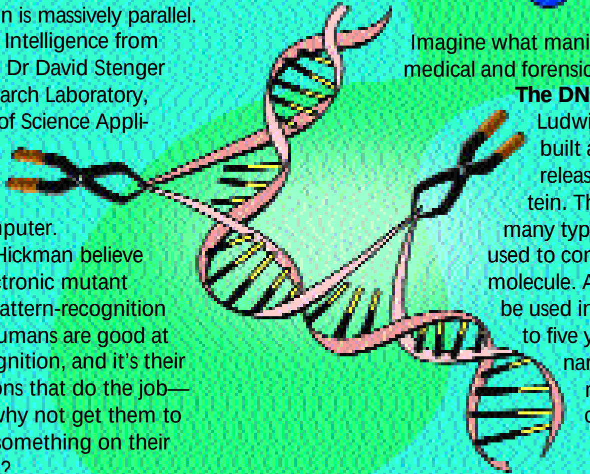
Nanodroplets: Researchers from the University of Texas M.D. Anderson Cancer Center created a programmable bio-chip that uses an array of electrodes to place water droplets on a surface, insert substances into them, and move and merge them as well. That doesn't sound great, but the 'droplets' can be 0.5 to 65 'nano'litres in volume. Applications are chemical and biological: The droplets can serve as carriers for cells, genes, and viruses.



Molecular Logic Gates: Researchers from the French National Center for Scientific Research (CNRS) and University College, London, devised a scheme to design logic circuits within individual molecules. It includes connecting a pair of benzene molecules to two gold electrodes, and the setup would be a real logic gate comprising two inputs and an output. The 1s and 0s here would be the rotational positions of the nitrogen and oxygen atoms in the benzene molecule.
Timeframe: 10 years



Magnetic Tweezers: Scientists at National Institute of Standards and Technology, based in Gaithersburg, Maryland, have demonstrated magnetic tweezers that can stretch, twist and uncoil **individual** bio-molecules such as strands of DNA. These work in conjunction with a special microscope. Imagine what manipulating individual molecules can do for bio-medical and forensic research!



The DNA Hand: In March this year, researchers from Ludwig Maximilians University (LMU) in Germany built a DNA molecular machine that binds to, and releases, **single molecules** of a specific type of protein. The DNA 'hand' can be made to select one of many types of protein, and at some point later on, be used to construct materials or machines, molecule by molecule. According to these researchers, the hand can be used in simple nano-construction applications in two to five years. This is reminiscent of the nanobots scenario—see the next section, 'Nanobots'—automated building of things molecule by molecule. Trust German engineering!